

TRIDENT

Triple-Redundant Integrated Design for Extraterrestrial Needs
A Maintainable Life-Support Focused Mars ISRU System

Status: Design Concept

Mass: 240–280 kg

Power: 850–1100 W

1. What TRIDENT Is

TRIDENT is a triple-redundant Mars In-Situ Resource Utilization (ISRU) architecture intended for early crewed surface operations. It prioritizes operational reliability and maintainability over maximum production rate. The system produces oxygen and methane from the Martian atmosphere using thermally integrated Sabatier and Solid Oxide Electrolysis Cell (SOEC) reactors.

Core design features:

- Three parallel reactor modules mounted on a shared thermal mass baseplate
- Warm-swap capability — one module can be isolated and partially cooled for maintenance while the remaining modules continue operating without thermal shock
- Parasitic RF-powered electrostatic precipitator for solid-state dust rejection (no consumable filters)
- Regenerative thermal coupling: Sabatier exothermic heat pre-heats SOEC water feedstock
- Optional ice-melt subsystem to close the hydrogen stoichiometric loop and produce bonus oxygen

2. What TRIDENT Is Not

TRIDENT is not a high-rate industrial propellant plant, not flight-qualified hardware, and not a plasma-core system. It is intentionally scoped as a life-support and limited-propellant node for early crews (approximately 4–7 people). An earlier, unrelated plasma-based concept (SPARK) has been publicly released separately and is not connected to this architecture.

3. Performance Summary

Oxygen Production	3.6 – 3.9 kg/sol	5.8 – 6.3 kg/sol
Methane Production	3.6 – 4.0 kg/sol	3.6 – 4.0 kg/sol
Continuous Power	850 – 1100 W	850 – 1100 W
Dry Mass	240 – 280 kg	240 – 280 kg
Crew Breathing Support	~4–5 people	~7 people

4. Architecture

Three identical reactor modules sit on a shared thermal mass structure. Each module contains a Sabatier reactor thermally coupled to an SOEC. The shared thermal mass stabilizes temperature and enables warm-swap maintenance: one module can be fluidically and thermally isolated and allowed to cool partially while the other two

continue production at operating temperature, avoiding thermal shock to the ceramic components.

Dust is managed by an RF-driven electrostatic precipitator powered by a parasitic tap from the reactor RF bus (typically 8–18 W). This eliminates the need for consumable HEPA-style filters in the dusty Martian environment.

The core loop (atmospheric CO₂ only) has a stoichiometric hydrogen deficit of 2 mol H₂ per mol CH₄ produced. An optional ice-melt subsystem uses waste heat to sublime Martian water ice and electrolyzes the resulting water, closing the hydrogen loop and producing additional oxygen.

5. Comparison with MOXIE

O ₂ Output	~0.25 kg/sol	5.8 – 6.3 kg/sol
Mass	17.1 kg	240 – 280 kg
Redundancy	Single-string	Triple + Shared Thermal Mass
Dust Mitigation	Filters	Parasitic RF ESP
Maintainability	None	Warm-swap capable
Primary Role	Technology demonstration	Life support + limited propellant

6. Performance Figures

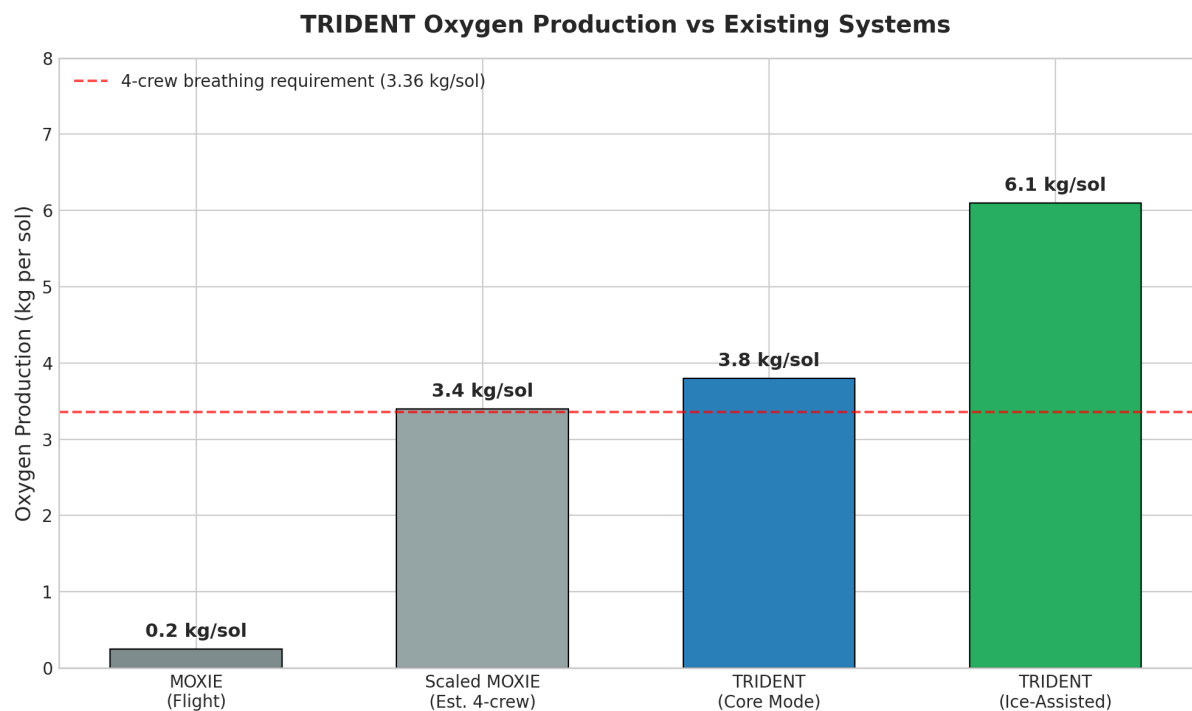


Figure 1 — Oxygen production comparison (kg/sol)

TRIDENT Mass Breakdown (Nominal 260 kg)

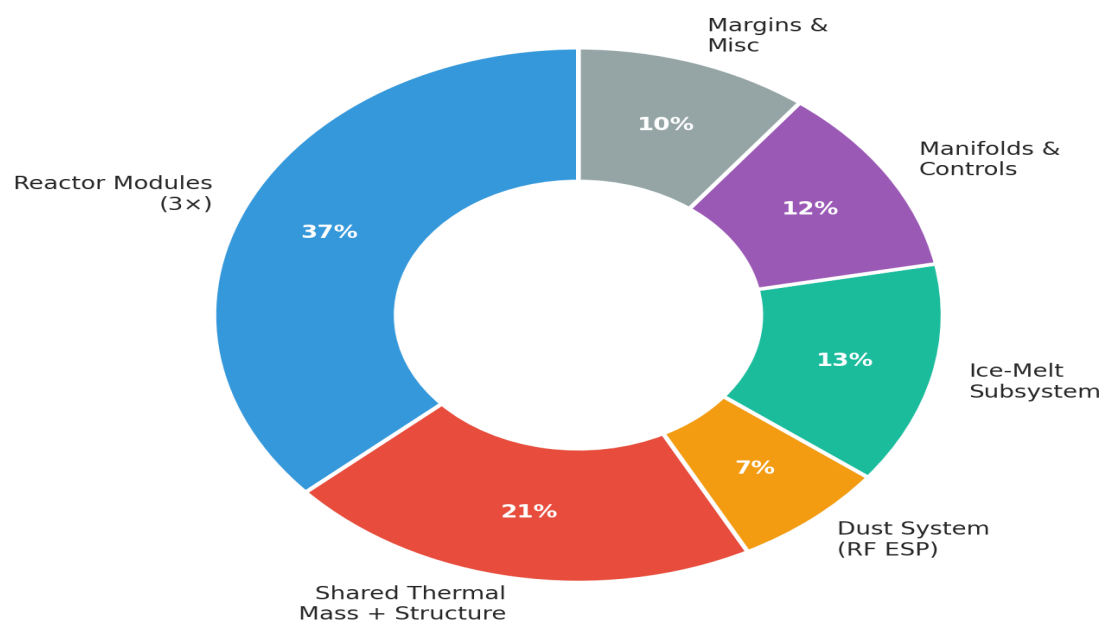


Figure 2 — Nominal mass breakdown (~260 kg)

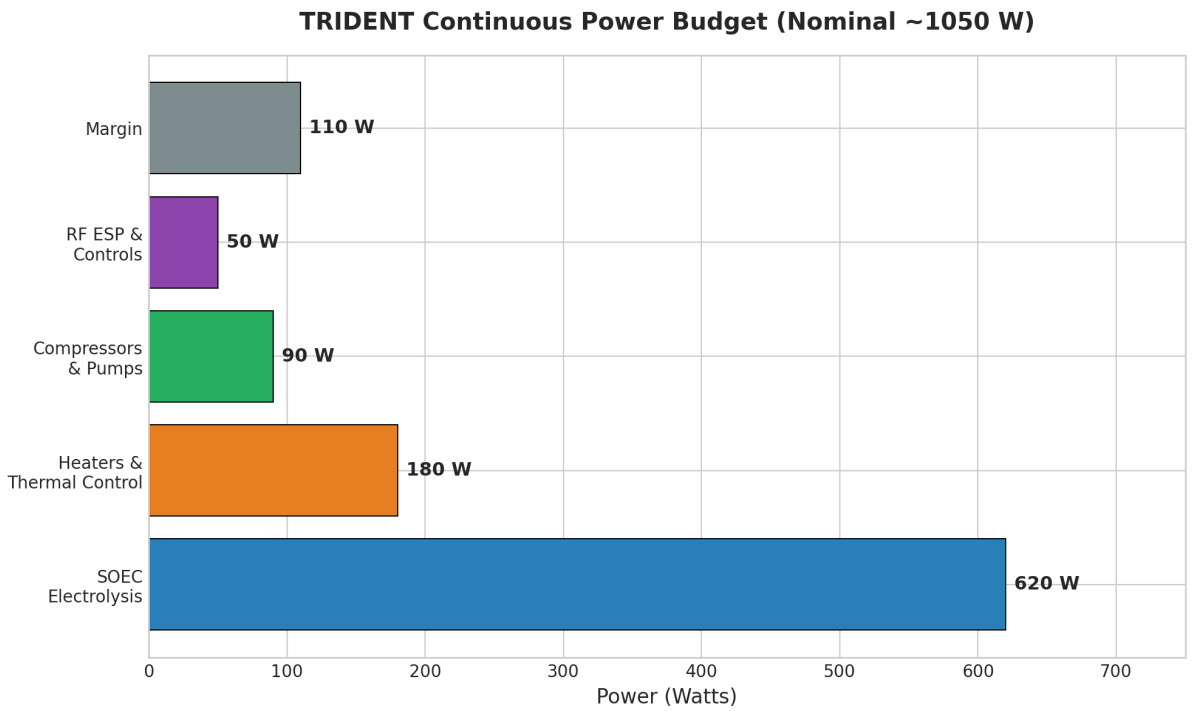


Figure 3 — Continuous power budget (nominal ~1050 W)

TRIDENT System Comparison

Metric	MOXIE (Flight)	TRIDENT (Core)	TRIDENT (Ice-Assisted)
O ₂ Output	~0.25 kg/sol	3.6–3.9 kg/sol	5.8–6.3 kg/sol
CH ₄ Output	None	3.6–4.0 kg/sol	3.6–4.0 kg/sol
Mass	17.1 kg	240–280 kg	240–280 kg
Power	~300 W peak	850–1100 W	850–1100 W
Redundancy	Single-string	Triple + Shared Thermal Mass	Triple + Shared Thermal Mass
Dust Mitigation	Filters	Parasitic RF ESP	Parasitic RF ESP
Maintainability	None	Warm-swap capable	Warm-swap capable
Primary Role	Technology demo	Life support + limited propellant	Life support + limited propellant

Figure 4 — System comparison summary

7. Installed Concept Mockup

The following concept render shows a TRIDENT unit installed on the Martian surface adjacent to a habitat module. Three cylindrical reactor modules sit on the shared thermal mass baseplate. The blue-glow intake represents the electrostatic dust precipitator. An astronaut is shown for scale.

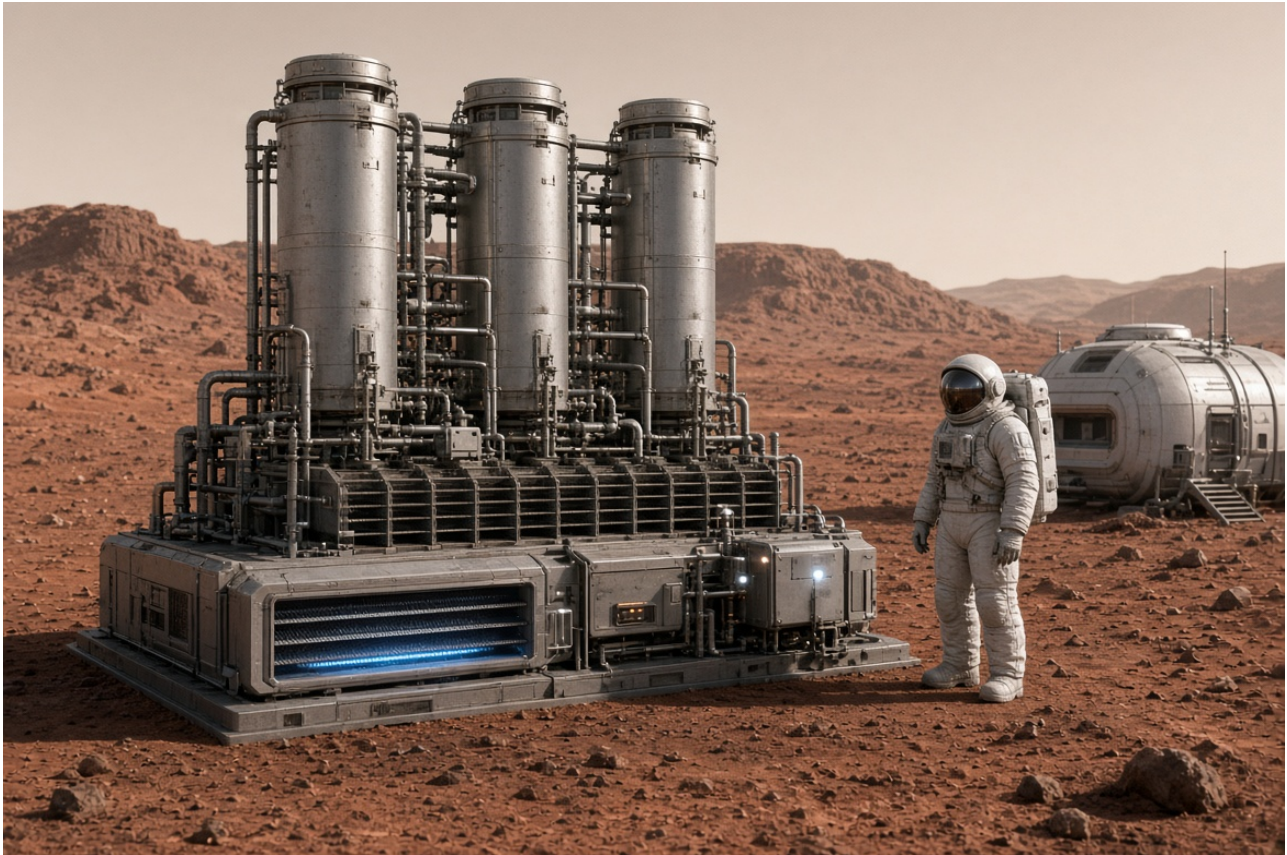


Figure 5 — TRIDENT installed concept (artist's rendering)

8. Design Philosophy & Honesty

TRIDENT was refined through multiple rounds of thermodynamic review. Earlier optimistic claims (significantly higher production rates, lower mass, true hot-swap of ceramic modules at full temperature, and fully passive thermoacoustic liquefaction) were deliberately corrected. The current specification uses conservative 2025–2026 engineering values for SOEC efficiency, Sabatier conversion, thermal losses, and mass estimation.

The system is intentionally scoped as a life-support and limited-propellant node rather than an industrial propellant plant. Gaseous storage is the baseline; active liquefaction is optional. The hydrogen stoichiometric deficit is openly acknowledged and solved only when the optional ice-melt subsystem is active.

9. Disclaimer

TRIDENT is a conceptual design for research, discussion, and engineering evaluation only. It is not flight hardware, has not been built or tested as a complete system, and must not be treated as a construction or operational plan. All performance figures are engineering estimates derived from publicly available data on SOEC, Sabatier, and thermal systems. Any real implementation would require independent detailed design, hazard analysis, qualification testing, and professional engineering oversight.